

Math 122: Exam 1 - Practice 2

Student Name: _____

Instructions:

- Show all work: Full credit requires complete, correct steps and proper math notation.
 - Points: Each question is worth 10 points.
 - Formula Sheet: Permitted (formulas only; no worked examples).
 - Calculators: Basic, non-graphing calculators allowed.
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Problem 1

Find the exact arc length of the curve over the interval from $x = 0$ to $x = 1$:

$$y = (2x + 1)^{3/2}$$

Problem 2

Find the limit:

$$\lim_{x \rightarrow 0} \frac{\tan(2x)}{\ln(1+x)}$$

Problem 3

Find the limit:

$$\lim_{x \rightarrow +\infty} x^2 e^{-2x}$$

Problem 4

Find the limit:

$$\lim_{x \rightarrow 0^+} (1 + 2x)^{1/x}$$

Problem 5Find $\frac{dy}{dx}$:

$$y = \cos^{-1} \left(\frac{1}{x^2} \right)$$

Problem 6

Find $\frac{dy}{dx}$:

$$y = \sin^{-1} x \cdot \cos^{-1}(x)$$

Problem 7

Evaluate the integral:

$$\int \frac{dx}{1 + 9x^2}$$

Problem 8

Evaluate the integral:

$$\int \frac{x \, dx}{\sqrt{1 - x^4}}$$

Problem 9

Evaluate the definite integral:

$$\int_0^1 \frac{(\sin^{-1} x)^3}{\sqrt{1-x^2}} dx$$

Problem 10

Evaluate the integral:

$$\int \frac{e^x}{1+e^{2x}} dx$$

— END OF EXAMINATION —